CENTRAL BOARD OF SECONDARY EDUCATION (CBSE)

ANNUAL EXAMINATION – 2025–26

SCIENCE (PHYSICS)

CHAPTER: SOUND

Class: IX
Time Allowed: 45 Minutes
Maximum Marks: 25

IMPORTANT NOTICE (READ CAREFULLY)

This question paper is the confidential property of the examination authority.

1. All candidates must read the instructions carefully before attempting the paper.
2. This question paper contains **5 questions**.
3. All questions are **compulsory**.
4. Marks allotted to each question are indicated against it.
5. Use of unfair means is strictly prohibited.
6. Draw neat and properly labeled diagrams wherever required.
7. Write answers clearly and legibly. Illegible answers may not be evaluated.
8. This document is **READ-ONLY**. Any modification, reproduction, or redistribution without permission is prohibited.

GENERAL INSTRUCTIONS

1. The question paper consists of **Section A to Section E**.
2. Attempt all questions in the space provided.
3. Use of calculators is not permitted.

SECTION A

Q1. Sound Propagation and Wave Properties (5 × 1 = 5 Marks)

Explain how sound is produced by a vibrating source and how it propagates through a medium to reach the observer. Define the following terms related to sound waves:

(a) Amplitude
(b) Frequency
(c) Wavelength
(d) Time Period

Answer Hint Box (For Examiner's Reference): - Must include **compressions** and **rarefactions**. - Must mention that sound is a **mechanical wave** and requires a **material medium**. - Correct definitions with **SI units** are compulsory. - Diagram of a longitudinal wave should be drawn.

SECTION B

Q2. The Physics of Echoes (4 Marks)

(a) What is an echo?
(b) Calculate the minimum distance required between a sound source and a reflector to hear a distinct echo at **20°C**. (Speed of sound in air = **344 m/s**)
(c) How will this minimum distance change if the temperature of air increases? Give reason.

Answer Hint Box (For Examiner's Reference): - Must mention **persistence of hearing (0.1 s)**. - Use formula: $2d = vt$. - Correct numerical calculation required. - Clear explanation of temperature effect.

SECTION C

Q3. Acoustic Engineering and Reverberation (5 Marks)

Define reverberation. Why is excessive reverberation undesirable in large auditoriums? Explain **two methods** used to control reverberation. Also state the purpose of using **curved ceilings** in concert halls.

Answer Hint Box (For Examiner's Reference): - Correct definition of reverberation. - Problems caused by excessive reverberation. - Use of sound-absorbing materials. - Explanation of curved ceilings for sound distribution.

SECTION D

Q4. SONAR and Ultrasound Applications (6 Marks)

(a) What is the full form of SONAR?
(b) Explain the working principle of SONAR with the help of a neat, labeled diagram to determine the depth of the sea.
(c) State **one industrial application** of ultrasound.

Answer Hint Box (For Examiner's Reference): - Correct expansion of SONAR. - Explanation must include **transmitter**, **receiver**, and **time measurement**. - Diagram is compulsory. - One correct industrial use.

SECTION E

Q5. Critical Thinking and Numerical Analysis (5 Marks)

Two children stand at opposite ends of a **1 km long aluminum rod**. One child strikes one end of the rod with a stone. The other child hears **two distinct sounds**.

(a) Explain the reason for hearing two sounds.
(b) Find the ratio of time taken by sound to travel through **air** and through **aluminum**.

(Given: Speed of sound in air = **346 m/s**; Speed of sound in aluminum = **6420 m/s**)

Answer Hint Box (For Examiner's Reference): - Sound travels at different speeds in different media. - Correct formula and calculation. - Final ratio must be clearly stated.

ANSWERS SECTION

(As per CBSE Annual Examination Marking Scheme)

Answer 1.

Sound is produced when an object vibrates. These vibrations cause the particles of the surrounding medium to vibrate, creating regions of **compressions** (high pressure) and **rarefactions** (low pressure). These pressure variations travel through the medium as a longitudinal mechanical wave and reach the observer. Sound requires a **material medium** and cannot travel in a vacuum.

- **Amplitude:** Maximum displacement of particles from their mean position. SI unit: meter (m).
- **Frequency:** Number of vibrations per second. SI unit: hertz (Hz).
- **Wavelength:** Distance between two consecutive compressions or rarefactions. SI unit: meter (m).
- **Time Period:** Time taken for one complete vibration. SI unit: second (s).

(A neat, labeled diagram of a longitudinal wave must be drawn.)

Answer 2.

(a) An echo is the repetition of sound heard after reflection from a distant surface.

(b) Due to **persistence of hearing**, the reflected sound must reach the ear after at least 0.1 s.

$$2d = vt$$

$$2d = 344 \times 0.1$$

$$d = 17.2\,m$$

(c) As temperature increases, the speed of sound increases. Hence, the **minimum distance required to hear an echo increases**.

Answer 3.

Reverberation is the persistence of sound in an enclosed space due to multiple reflections from walls, ceilings, and floors. Excessive reverberation causes overlapping of sounds, making speech unclear.

It can be controlled by: 1. Using sound-absorbing materials like carpets, curtains, fibreboards. 2. Providing false ceilings and acoustic panels.

Curved ceilings help in uniform distribution of sound waves and prevent dead spots in auditoriums.

Answer 4.

(a) SONAR stands for **Sound Navigation And Ranging**.

(b) SONAR works on the principle of reflection of ultrasonic waves. A transmitter sends ultrasonic waves towards the seabed. These waves are reflected back and received by a detector. The time interval between transmission and reception is used to calculate the depth of the sea.

(Diagram showing ship, transmitter, receiver, and wave paths must be drawn.)

(c) Ultrasound is used for detecting cracks in metal blocks.

Answer 5.

(a) Two sounds are heard because sound travels faster through solid aluminum than through air. The sound through the rod reaches earlier.

(b)

$$t_{air} = 1000/346$$

$$t_{aluminum} = 1000/6420$$

Ratio:

$$t_{air} : t_{aluminum} = 6420 : 346 \approx 18.55 : 1$$

End of CBSE Question Paper & Answers